THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Methods of Separation in Everyday Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture of iron filings, common salt, and fine sand is given. Which sequence is the most efficient? 1

(a) add water, evaporate, magnet
(b) evaporate first, filter, magnet
(c) magnet, add water and filter, evaporate filtrate
(d) filter, magnet, sieve

2. 100 g of a mixture contains 20 g of iron, 30 g of sand and 50 g of salt. After separation, the mass of sand obtained is expected to be: 1

(a) 30 g **(b)** 100 g **(c)** 50 g **(d)** 20 g

3. Why is alum added during water purification before filtration? 1

(a) it makes tiny particles clump and settle **(b)** it kills all germs
(c) it adds taste **(d)** it dissolves salt

4. In the mixture above, after the first step the residue on the filter paper is: 1

(a) water **(b)** salt **(c)** sand **(d)** iron

5. A saturated solution of salt at 30 °C contains 36 g salt in 100 g water. In 50 g water, the maximum salt that dissolves is: 1

(a) 36 g **(b)** 18 g **(c)** 9 g **(d)** 72 g

6. Why does purifying water require both filtration and chlorination or boiling? 1

(a) boiling removes sand **(b)** sedimentation removes germs
(c) filtration cannot remove all germs **(d)** filtration removes salt

7. Which of these will take the longest to settle in water? 1

(a) pebbles **(b)** sand **(c)** gravel **(d)** fine clay

8. Using the information above, if 25 g of salt is added to 50 g water at 30 °C, the extra undissolved salt is: 1

(a) 18 g **(b)** 25 g **(c)** 0 g **(d)** 7 g

9. **Assertion (A):** Winnowing is used to separate heavier grains from lighter husk. 1
Reason (R): Wind blows away the lighter husk while the heavier grains fall down.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A mixture of sand and water can be separated by filtration. 1
Reason (R): Salt can be obtained from sea water by evaporation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A mixture of 50 g salt and 50 g sand is given. After separation, 49 g of salt and 50 g of sand are recovered. Explain the 1 g loss. 2

12. How would you separate a mixture of sugar crystals and wheat grains? 2

13. How do we remove insoluble impurities from honey? 2

14. Why does tea powder pass through a sieve but not through a filter paper in tea? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture of salt, sand, chalk powder, iron filings and wooden chips needs to be separated. Suggest the best sequence. 3

16. Describe why a mixture of water and kerosene separates into two layers. 3

17. How does a cotton cloth help separate curd from whey? 3

OR

At 20 °C, 36 g of salt dissolves in 100 g of water. How much water is needed to dissolve 90 g of salt? 3

18. Why is a cloth filter used when preparing curd or paneer? 3

OR

A mixture of oil and sand is given. How will you separate? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** After harvest, farmers in Punjab beat the wheat stalks to loosen the grains. Then they pour the grain and husk from a height in the wind, and the husk blows away to a side. The cleaned grain is later passed through a sieve to remove small stones. 5
(i) Name the process of beating the stalks.
(ii) Name the process using wind.
(iii) Which method removes small stones?
(iv) Why is winnowing not suitable for separating sand from water?

20. Write a step-wise plan to obtain pure water, pure salt and pure sand from sea-side sandy salt water. 5

OR

Describe the separation of the following mixture: rice grains, husk, small stones, iron pieces. 5

END OF QUESTION PAPER — Total: 40 Marks